

Dylan DiBeneditto

Electrical Engineering

[dylandibeneditto.github.io](https://github.com/dylandibeneditto) dyland26@vt.edu

Education

Virginia Tech

B.S. Electrical Engineering

Expected 2030

Skills

Hardware / EE PCB/Schematic Design (KiCad), STM32/ESP32/nRF board design, sensor integration (IMU, barometer, magnetometer), Bluetooth/Wifi (2.4GHz) RF PCB antenna design, JLCPCB Manufacturing (Gerber, BOM, CPL files), power and LiPo charging circuit design.

Software Python, C, C++, Javascript/TypeScript, Java, Go, Rust, Lua, Swift, embedded programming, frameworks like React, NextJS, Svelte/SvelteKit, SwiftUI.

Tools KiCad, Git, Fusion360, JLCPCB/PCBA Toolchain

Electrical and Embedded Projects

K42

2026

Independent Hardware Product

Designed a 42-key wireless split ergonomic keyboard around a bare-die nRF52840 (QFN-73) MCU, with multi-sheet KiCad schematics, DRV2605L + LRA haptic feedback, OLED displays, and Kailh Choc hotswap sockets. Planned full JLCPCB PCBA production and evaluated a direct to consumer launch strategy (via a Stripe + NextJS Web App) with margin modeling. Ultimately costs by tariffs resulting in simplification of the board to allow for simpler hand soldering.

Null36

2025-2026

Independent Hardware Product

My first PCB design, used a module for the nRF52840 MCU and designed a key matrix and layout in order to fit the shape of my hand and alleviate hand fatigue from programming without spending \$300+ on a commercial board.

Software and Creative Projects

'fmov' Python Generation Video Engine

2025

12,000+ downloads

Developed a Python-based video generation engine that interfaces with the lower level C-based library Ffmpeg. This allows for simple video creation without having to leave python, making it the most performant solution for video generation in Python.

Waldron Catering

2025

Paid Full-Stack Web Developer and Graphic Designer

Designed the brand identity for a catering business, built the website, deployed and maintained it. The website uses HTML, CSS, and JavaScript and is highly optimized for all screen sizes and SEO.

Neural Network in a Calculator

2025

Winner, Global Desmos Art Contest (top 15 in age group)

Designed a multi-layer perceptron neural network for a web-based calculator in order to recognize handwritten digits. Self-trained the model using data from the MNIST dataset.

'PrintJava', Programmatical 3D Printing

2024-2026

Highschool Project, now part of CS curriculum

Designed a Java library as president of my school's CS National Honor Society to receive a grant for a 3D printer. The library allowed for the programmatic creation of intricate 3D models from spheres to 4D fractal objects. Near the end of my senior year I was able to design a curriculum around the library for future students to learn from.